HARIRAMAKRRISHNAN RAMACHANDRAN

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PROFESSIONAL EXPERIENCE — 3.5 YEARS

Software Engineer — HCL Technologies, Chennai, India Sep 2021 – Jan 2025

- Developed and maintained **Linux kernel** components including storage drivers and block drivers for modem processors and SoCs.
- Optimized **storage and memory performance** for runtime efficiency and power consumption across flash technologies.
- Built and maintained **Linux user-space tools** for storage testing and validation, including image creation utilities.
- Debugged and resolved **kernel-level issues** reported by customers and flash vendor partners during commercialization.
- Contributed to upstream **Linux kernel development** and engaged with open-source communities for new feature implementation.
- Wrote **Python scripts** to automate flash provisioning tasks for development and factory environments, reducing manual effort by **~40%**.
- Containerized storage driver development environments using **Docker**, enabling consistent build and test setups across developer machines.

SKILLS

Embedded C/C++ – C, Embedded C, Real-time Operating Systems (RTOS), Device Drivers, Kernel Internals, Memory Management, Scheduling, Driver Frameworks

Linux Kernel Development – Linux Kernel, Storage Drivers, Filesystem Modules, Block Drivers, User Space Tools, Debugging (JTAG/ICE)

Storage Technologies – UFS, NVMe, SD Express, eMMC, NAND, NOR, FAT, GPT, LittleFS, F2FS, EXT4, EROFS

Programming & Scripting – Python, ARM Architecture

Development Tools & Practices – Git, Jenkins, CI/CD, Docker, Agile Methodologies, Code Reviews

PROJECTS

Linux Kernel Storage Driver Development

- Engineered a new **Linux kernel driver** for a custom flash storage controller, focusing on high-throughput data transfer.
- Implemented **filesystem module support** for FAT and LittleFS, ensuring compatibility with various embedded systems.
- Developed **user-space utilities in Python** for testing driver stability and performance under simulated load conditions, achieving **~5,000 IOPS**.

Embedded Software for Flash Provisioning Tools

- Developed a **PC tool in C** for creating bootable images (GPT format) for development boards.
- Created a **storage flashing utility** that supports UFS devices, used in both development and factory pre-production.
- Integrated **error handling and reporting** mechanisms to provide detailed feedback during flash provisioning processes.

EDUCATION

M.Sc. Data Analytics — National College of Ireland, Dublin Feb 2026

B.E. Computer Science — SNS College of Technology, Coimbatore, India 2021

CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)